

BEYOND THE FORD-CIRCLE LAW: PARABOLIC RENORMALIZATION AND THE SHAPE OF MANDELBROT SATELLITE BULBS

LARS BENDIXEN

ABSTRACT. To first order, the satellite component of the Mandelbrot set attached to the main cardioid at internal angle p/q is a round disc of diameter proportional to q^{-2} ; this is the Ford-circle law. We study the first correction. Expanding the multiplier of the bifurcating q -cycle in the bulb-scale parameter $u = q^2\varepsilon$, we prove that the first coefficient depending on p/q is $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$, where $\iota_{p/q}$ is the holomorphic index of the parabolic fixed point of f^q . We also prove that $\kappa(p/q) + \frac{1}{2}$ is the index of the rotation quotient of f^q , and we give an exact $O(q^2)$ recursion for it. We then present numerical evidence, at up to 27 digits, that the correction is governed by parabolic renormalization. The limit $\kappa_0 = \lim \kappa(1/q)$ equals $\iota(\mathcal{P}_0) - \frac{1}{2}$, where $\mathcal{P}_0$ is the parabolic renormalization of $z + z^2$. Along bounded p , the limits of κ and of the bulb diameter are those of $\mathcal{P}_0$ times the Inou-Shishikura phase. The entire $O(1/q)$ correction is kinematic, because the exact phase is a Möbius function of u . Along continued fractions with large partial quotients the constants form a hierarchy that converges geometrically, with ratio 0.042, to a constant κ_* attached to the fixed point of parabolic renormalization composed with complex conjugation, which is not the Inou-Shishikura fixed point. Components differ only through the germ at their root: the period-two disc is governed by the renormalization of $-z + z^2$. The cubic family $z^3 + c$ forms a second universality class. We compute three new constants to 30 digits, and no integer relation with classical constants is found.

1. INTRODUCTION

1.1. The Ford-circle law and its first correction. Let $\lambda \in \mathbb{D}$ parametrize the main cardioid of the Mandelbrot set M by the multiplier of the attracting fixed point, $c(\lambda) = \lambda/2 - \lambda^2/4$. Equivalently we work with

$$f_\lambda(z) = \lambda z + z^2,$$

which is affinely conjugate to $z^2 + c(\lambda)$. At each root of unity $\lambda_0 = e^{2\pi i p/q}$, $\gcd(p, q) = 1$, a satellite hyperbolic component $W_{p/q}$ of period q is attached to the cardioid [2]. Its size is of order q^{-2} [7]. More precisely, to first order in $1/q$ it is a round disc of diameter $2|c'(\lambda_0)|q^{-2} = 2\sin(\pi p/q)q^{-2}$ [5]. These diameters and the tangency pattern of the $W_{p/q}$ are those of the Ford circles [6], which is the leading-order content of the familiar analogy between M and the Farey tessellation.

This paper studies the *first correction* to that law, which to our knowledge has not been investigated before. Put $\lambda = \lambda_0 e^\varepsilon$ and let $\rho_{p/q}(\varepsilon)$ be the multiplier of the q -cycle of f_λ that bifurcates from the fixed point 0. In the bulb-scale variable $u = q^2\varepsilon$ write

$$(1) \quad R_{p/q}(u) := \rho_{p/q}(u/q^2) = \sum_{k \geq 0} r_k(p/q) u^k.$$

Date: September 2026.

2020 Mathematics Subject Classification. Primary 37F46; Secondary 37F25, 37F10, 11A55.

Key words and phrases. Mandelbrot set, satellite components, Ford circles, holomorphic index, résidu itératif, parabolic implosion, horn map, near-parabolic renormalization.

Up to the conformal change of variable $u \mapsto c(\lambda_0 e^{u/q^2})$, the component $W_{p/q}$ is the set $\{|R_{p/q}(u)| < 1\}$. We show that $r_0 = 1$ and $r_1 = -1$ for every p/q , so that to first order $W_{p/q}$ is the disc $|u - 1| < 1$: this is the Ford law. The first coefficient that depends on p/q is

$$\kappa(p/q) := r_2(p/q),$$

and it governs the leading deviation of $W_{p/q}$ from a round disc.

1.2. Exact results. Let $f = f_{\lambda_0}$, so that $f^q(z) = z + a z^{q+1} + O(z^{q+2})$ with $a \neq 0$, and let $\iota_{p/q} = \text{Res}_{z=0} dz/(z - f^q(z))$ be the holomorphic index of 0 as a fixed point of f^q .

- **Index formula** (Theorem 2.1). $\rho_{p/q}(\varepsilon) = 1 - q^2\varepsilon + q^3(\iota_{p/q} - \frac{1}{2})\varepsilon^2 + O(\varepsilon^3)$, that is, $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$. In particular $\kappa(p/q) \in \mathbb{Q}(\zeta_q)$, and the values $\kappa(p'/q)$, $\gcd(p', q) = 1$, form a single Galois orbit. The same holds for the satellites of every hyperbolic component of $z^d + c$.
- **Normal form** (Theorem 3.1, Corollary 3.2). If $N(z) = \lambda_0 z(1 + b_1 z^q + b_2 z^{2q})$ is the resonant normal form of f , then $q \iota_{p/q} = \frac{q^2-1}{2} + b_2/b_1^2$ and $\kappa(p/q) = \iota(\Phi_{p/q}) - \frac{1}{2}$, where $\Phi_{p/q}(W) = N^q(W^{1/q})^q$ is the q -fold rotation quotient of f^q . The coefficients b_1, b_2 come from an explicit recursion with $O(q^2)$ operations.
- **Horn-map indices** (Theorem 4.3). If $E(Z) = Z + a_0 + \sum_{n \geq 1} a_n e^{2\pi i n Z}$ is an upper horn map and $\mathcal{P}(W) = W \exp(2\pi i \sum_n a_n W^n)$ the corresponding parabolic germ, then $\iota(\mathcal{P}) - \frac{1}{2} = a_2/(2\pi i a_1^2)$, independently of all normalizations.
- **Kinematics** (Proposition 5.1). If $\lambda = e^{2\pi i \alpha} = \lambda_0 e^{u/q^2}$, then exactly $e^{-2\pi i/\alpha} = e^{-2\pi i q/p} e^{\tilde{u}/p^2}$, where $\tilde{u} = u/(1 + \frac{u}{2\pi i p q})$ is a Möbius function of u .

1.3. Numerical results and conjectures. The rest of the paper supports one picture. The rotation quotient $\Phi_{p/q}$ is, up to $O(q^{-2})$, a near-parabolic renormalization of $f_{p/q}$ in the sense of Inou and Shishikura [8], composed with the exact phase of Proposition 5.1. The main consequences, each tested against the bulbs, are listed below. Table 1 collects the constants.

- (1) *The constant κ_0* (§4). $\kappa(1/q) \rightarrow \kappa_0 = \iota(\mathcal{P}_0) - \frac{1}{2}$, where $\mathcal{P}_0$ is the parabolic renormalization of $z + z^2$. Bulbs certified to $q = 32768$ and the horn map agree to $3 \cdot 10^{-28}$.
- (2) *Bounded p and the bulb size* (§5). Along $q = pN + r$ the limits of κ and of the bulb diameter are those of the family $e^{-2\pi i r/p} e^{u/p^2} \mathcal{P}_0$. In ten classes with $p \leq 5$ they agree to at most $2.6 \cdot 10^{-8}$ for κ and $9 \cdot 10^{-7}$ for the diameter.
- (3) *Kinematics* (§5.4). $R_{p/q}(u) = \rho_{p,r}(\tilde{u}(u)) + O(q^{-2})$ coefficientwise. This accounts for the universal term $-i/(2\pi p q)$ in κ and for the whole $O(1/q)$ part of $R_{p/q}$.
- (4) *The hierarchy* (§6). Iterating along continued fractions with large partial quotients gives constants κ_k that contract by the factor 0.042 per level towards κ_* . The levels $k \leq 3$ are confirmed on bulbs. The fixed point differs from that of Lanford and Yampolsky [10], whose computations our code reproduces.
- (5) *Component dependence* (§7). The bounded- p limits of a component depend only on the parabolic germ at its root. For the period-two disc this germ is $-z + z^2$, with its own constant $\kappa_{1/2}$.
- (6) *Degree three* (§8). For $z^3 + c$ the same statements hold with the cusp germ $v + v^2 + \frac{1}{3}v^3$; the bulbs now have two antipodes (Proposition 8.1).

We state as conjectures only what the evidence supports quantitatively: Conjectures 4.5, 5.2, 6.3 and 7.1. All numerical results are obtained either in certified ball arithmetic or by two independent computations, and they are stated with their error estimates (Appendix A).

constant	germ	value	bulbs	
κ_0	$z + z^2$	$0.02382588740220057 - 0.05230465911400367 i$	$3 \cdot 10^{-28}$	§4
$\kappa_0^{(3)}$	$v + v^2 + \frac{1}{3}v^3$	$0.1435992780952916 - 0.03031164789850254 i$	$1.3 \cdot 10^{-6}$ ($q = 1024$)	§8
$\kappa_{1/2}$	$-z + z^2$	$0.01840526161796478 - 0.04290372388023048 i$	$3 \cdot 10^{-10}$	§7
κ_*	$G_* = \mathcal{P}(\bar{G}_*)$	$0.011929183413 - 0.037455493752 i$	levels ≤ 3	§6
G_0	$e^u \mathcal{P}_0$	1.02641539	$4 \cdot 10^{-8}$	§5.3

TABLE 1. The constants of this paper. The column “bulbs” gives the agreement between the horn-map value and the limit extracted from the bulbs; for $\kappa_0^{(3)}$ it gives the error at $q = 1024$, which decays like q^{-2} .

1.4. Related work. The q^{-2} law is due to Guckenheimer and McGehee [7], and the leading constant $2 \sin(\pi p/q)$ was derived by Fowler and McGuinness [5]. Parabolic implosion and horn maps go back to Douady, Lavaurs and Shishikura [11, 16]. Near-parabolic renormalization and its invariant class are due to Inou and Shishikura [8], and Lanford and Yampolsky [10] studied the fixed point of parabolic renormalization numerically. The holomorphic index and the résidu itératif are classical [12, 3, 1]. The Ford-circle analogy is usually stated only to leading order, and we are not aware of previous work on the corrections.

1.5. Organization. Section 2 proves the index formula, and Section 3 gives the normal form and the rotation quotient. Section 4 treats the horn map of $z + z^2$ and κ_0 . Section 5 formulates the renormalization principle and tests it at bounded p and at first order in $1/q$. Sections 6 and 7 treat the hierarchy and the dependence on the component, and Section 8 treats degree three. Section 9 reports integer-relation searches, and Section 10 lists open problems. The appendix describes the numerical methods.

2. THE INDEX FORMULA

Fix p/q in lowest terms, $\lambda_0 = e^{2\pi i p/q}$, and write $f = f_{\lambda_0}$. Since f has a single critical point, the parabolic fixed point 0 has exactly one cycle of q attracting petals, so

$$(2) \quad f^q(z) = z + a z^{q+1} + O(z^{q+2}), \quad a = a(p/q) \neq 0.$$

The holomorphic index [12, §12] of 0 as a fixed point of f^q is

$$\iota_{p/q} := \frac{1}{2\pi i} \oint \frac{dz}{z - f^q(z)} = \text{Res}_{z=0} \frac{dz}{z - f^q(z)},$$

and its résidu itératif [3, 1] is $\text{résit} = \frac{q+1}{2} - \iota_{p/q}$.

Theorem 2.1 (Index formula). *The multiplier $\rho_{p/q}(\varepsilon)$ of the q -cycle of $f_{\lambda_0 e^\varepsilon}$ that bifurcates from 0 is holomorphic near $\varepsilon = 0$, and*

$$\rho_{p/q}(\varepsilon) = 1 - q^2 \varepsilon + q^3 \left(\iota_{p/q} - \frac{1}{2} \right) \varepsilon^2 + O(\varepsilon^3).$$

Equivalently, in (1), $r_0 = 1$, $r_1 = -1$ and $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$.

Proof. Choose a disc D around 0 on whose boundary $z \neq f^q(z)$ and which contains no other fixed point of f^q . By (2) and Rouché’s theorem, for $|\varepsilon|$ small the map f_λ^q , $\lambda = \lambda_0 e^\varepsilon$, has exactly $q + 1$ fixed points in D , counted with multiplicity. One of them is 0, with multiplier $\lambda^q = e^{q\varepsilon} \neq 1$, so it is simple. For each proper divisor d of q the map f^d has multiplier $\lambda_0^d \neq 1$ at 0, so f_λ^d has no fixed point in D other than 0. The remaining q points therefore have exact period q and form a single cycle; call its multiplier $\rho(\varepsilon)$. The cycle, as an unordered set,

depends holomorphically on λ , hence so does ρ , which is a symmetric function of the cycle points. The function ρ is not identically 1, since the cycle is attracting on the satellite, so for $\varepsilon \neq 0$ small all $q+1$ fixed points are simple. Summing their indices $1/(1-m)$ gives

$$(3) \quad \frac{1}{1-e^{q\varepsilon}} + \frac{q}{1-\rho(\varepsilon)} = \frac{1}{2\pi i} \oint_{\partial D} \frac{dz}{z-f_\lambda^q(z)} \xrightarrow{\varepsilon \rightarrow 0} \iota_{p/q},$$

because the integrand converges uniformly on ∂D . The first term equals $-1/(q\varepsilon) + \frac{1}{2} + O(\varepsilon)$ and is unbounded. Hence $\rho(0) = 1$. Writing $1-\rho = A\varepsilon^m(1+O(\varepsilon))$ with $A \neq 0$, boundedness of the sum forces $m = 1$ and $A = q^2$. With $\rho = 1 - q^2\varepsilon + C\varepsilon^2 + O(\varepsilon^3)$ we get $q/(1-\rho) = 1/(q\varepsilon) + C/q^3 + O(\varepsilon)$, and (3) yields $\frac{1}{2} + C/q^3 = \iota_{p/q}$. $\square$

Remark 2.2 (Other components). The proof uses only that the parabolic point has multiplicity $q+1$. It therefore applies verbatim to the satellites of any hyperbolic component of $z^d + c$, whose parabolic cycles have a single cycle of petals, with λ the multiplier of the attracting cycle of that component and f^q replaced by the first return map. We checked numerically, for the period-two disc and for $z^3 + c$, that $r_2 = (\iota - \frac{1}{2})/q$ holds to 10^{-7} at $p/q = 2/7$ and $3/11$.

Remark 2.3 (Exact values). Since f^q has coefficients in $\mathbb{Z}[\lambda_0]$, the index is a rational function of λ_0 with rational coefficients. Hence $\iota_{p/q} \in \mathbb{Q}(\zeta_q)$ and $\iota_{p'/q} = \sigma(\iota_{p/q})$ for the automorphism $\sigma : \zeta_q \mapsto \zeta_q^{p'\bar{p}}$, where $pp \equiv 1 \pmod{q}$. For example $\iota_{1/2} = \frac{1}{8}$, $\iota_{1/3} = \frac{100-8\sqrt{3}i}{441}$ and $\iota_{1/4} = \frac{1447-365i}{4624}$, so that $\kappa(1/2) = -\frac{3}{16}$ and $\kappa(1/3) = \frac{-241-16\sqrt{3}i}{2646}$.

Definition 2.4. Let $c_{\text{root}} = c(\lambda_0)$, and let $c_{\text{ant}} \in \partial W_{p/q}$ be the boundary point at which the q -cycle has multiplier -1 . It is unique because in degree two the multiplier map $W_{p/q} \rightarrow \mathbb{D}$ is a conformal isomorphism. Set

$$d(p/q) = |c_{\text{ant}} - c_{\text{root}}|, \quad G(p/q) = \frac{d(p/q)}{2|c'(\lambda_0)|q^{-2}}.$$

Since $c(\lambda_0 e^{u/q^2}) - c(\lambda_0) = c'(\lambda_0)\lambda_0 u/q^2 + O(q^{-4})$, we have $G(p/q) = \frac{1}{2}|u_a| + O(q^{-2})$, where $R_{p/q}(u_a) = -1$. To first order $u_a = 2$ and $G = 1$, and the deviation of G from 1 is governed at leading order by κ .

3. NORMAL FORMS AND THE ROTATION QUOTIENT

Theorem 3.1 (Normal form). *There is a unique formal conjugacy $H(z) = z + \sum_{k \geq 2} H_k z^k$ with $H_{q+1} = H_{2q+1} = 0$ such that $f \circ H = H \circ N$ up to order $2q+1$, where*

$$N(z) = \lambda_0 z(1 + b_1 z^q + b_2 z^{2q}).$$

Put $\beta_i = \lambda_0 b_i$ and let $(H^2)_k$ denote the coefficient of z^k in $H(z)^2$. The coefficients are determined by the recursion

$$(4) \quad H_k(\lambda_0 - \lambda_0^k) = [k > q+1] (k-q)\lambda_0^{k-q-1}\beta_1 H_{k-q} - (H^2)_k \quad (2 \leq k \leq 2q, k \neq q+1),$$

together with $\beta_1 = (H^2)_{q+1}$ and $\beta_2 = (H^2)_{2q+1}$. Moreover

$$a(p/q) = q b_1, \quad q \iota_{p/q} = \frac{q^2 - 1}{2} + \frac{b_2}{b_1^2}, \quad \kappa(p/q) = \frac{1}{2} - \frac{1}{2q} - \frac{1}{2q^2} + \frac{b_2}{(qb_1)^2}.$$

Proof. Comparing coefficients of z^k in $\lambda_0 H + H^2 = \sum_m H_m N^m$ gives (4). Here we use that $[N^m]_k$ is nonzero only for $k \in \{m, m+q, m+2q\}$, with values λ_0^m and $m\lambda_0^{m-1}\beta_1$ in the first two cases. The factor $\lambda_0 - \lambda_0^k$ vanishes exactly when $k \equiv 1 \pmod{q}$. At $k = q+1$ and $k = 2q+1$ the

equation fixes β_1 and β_2 and leaves H_{q+1} and H_{2q+1} free; we set them to zero. For the second part, write $N^j(z) = \lambda_0^j z(1 + A_j z^q + B_j z^{2q} + \dots)$. Using $\lambda_0^q = 1$ and $(N^j)^q = z^q(1 + qA_j z^q + \dots)$,

$$A_{j+1} = A_j + b_1, \quad B_{j+1} = B_j + (q+1)b_1 A_j + b_2,$$

so $A_q = qb_1$ and $B_q = \frac{(q+1)q(q-1)}{2}b_1^2 + qb_2$. The coefficient a and the index are invariant under formal conjugacy tangent to the identity, and for $z + Az^{q+1} + Bz^{2q+1}$ the index is B/A^2 . This gives the formulas for a and ι , and the formula for κ follows from Theorem 2.1. $\square$

Corollary 3.2 (κ is an index). *Since N commutes with $z \mapsto \lambda_0 z$, the rotation quotient $\Phi_{p/q}(W) := N^q(W^{1/q})^q = W + q^2 b_1 W^2 + \dots$ is a well-defined germ, and*

$$\kappa(p/q) = \iota(\Phi_{p/q}) - \frac{1}{2} = \frac{1}{2} - \text{résit}(\Phi_{p/q}).$$

Proof. From $N^q(z) = z(1 + A_q z^q + B_q z^{2q} + \dots)$ we get $\Phi(W) = W(1 + A_q W + B_q W^2)^q + \dots$. Hence $\Phi = W + AW^2 + BW^3 + \dots$ with $A = qA_q = q^2 b_1$ and $B = qB_q + \binom{q}{2}A_q^2$. Then $\iota(\Phi) = B/A^2 = 1 - \frac{1}{2q} - \frac{1}{2q^2} + \frac{b_2}{(qb_1)^2}$, which is $\kappa + \frac{1}{2}$ by Theorem 3.1. $\square$

Remark 3.3 (Computation). The recursion (4) costs $O(q^2)$ arithmetic operations. It is numerically stable: at $q = 128$ and $q = 301$ the computed $\kappa(1/q)$ does not change in more than twenty digits when the working precision is raised from 60 to over 300 digits, and for generic p with q up to $6.4 \cdot 10^4$ two runs at 128 and 192 bits agree to 10^{-34} . By contrast, composing f symbolically q times costs $O(q^3)$ and loses about q decimal digits to cancellation at $p = 1$, where $|a(1/q)| \approx e^{2.3q}$. With a blocked online convolution the recursion runs in ball arithmetic [9], which certifies $\kappa(1/q)$ for $q \leq 32768$ (Appendix A).

4. THE HORN MAP OF $z + z^2$ AND THE CONSTANT κ_0

4.1. The limit along $1/q$. Table 2 and Figure 1 show $\kappa(1/q)$ for q up to 32768.

Numerical result 4.1. Least-squares fits $\kappa(1/q) = \sum_{j \leq J} c_j q^{-j}$ over $q \geq q_{\min}$ are stable under changes of $(q_{\min}, J)$. The best overdetermined fit, $(q_{\min}, J) = (256, 10)$ on thirteen values with residual $3.6 \cdot 10^{-29}$, gives

$$c_0 = 0.023825887402200569000729189645 \dots - 0.052304659114003666316522486208 \dots i,$$

the first-order coefficient $c_1 = -i/(2\pi)$ to within $2 \cdot 10^{-23}$, and $c_2 = 0.402776138694 + 2.902726170410i$. The ratios $|c_{j+1}/c_j|$ stay near 7, consistent with an expansion that converges for $q \gtrsim 7$.

4.2. Fatou coordinates and the horn map. We recall the objects of parabolic implosion [11, 16, 8, 10] in the form in which we compute them. In the coordinate $w = -1/z$ the map $f_0(z) = z + z^2$ becomes

$$F(w) = \frac{w^2}{w-1} = w + 1 + \frac{1}{w-1}.$$

Lemma 4.2 (Formal Fatou coordinate). *There is a unique formal series $\Phi(w) = w - \log w + \sum_{k \geq 1} c_k w^{-k}$ with $\Phi \circ F = \Phi + 1$. Its coefficients are rational,*

$$c_1, c_2, \dots = \frac{1}{2}, \frac{1}{3}, \frac{13}{36}, \frac{113}{240}, \frac{1187}{1800}, \frac{877}{945}, \dots,$$

and with $u = 1/w$ they satisfy the triangular system

$$\sum_{k \geq 1} c_k [(u - u^2)^k - u^k] = - \sum_{m \geq 2} \left(1 - \frac{1}{m}\right) u^m.$$

q	$\kappa(1/q)$	$ \kappa(1/q) + \frac{i}{2\pi q} - \kappa_0 $
64	0.02385454027565757 – 0.05408067667475267 i	$7.1 \cdot 10^{-4}$
256	0.02383094323212187 – 0.05288194473715776 i	$4.5 \cdot 10^{-5}$
1024	0.02382625451962614 – 0.05245731336604759 i	$2.8 \cdot 10^{-6}$
4096	0.02382591114406885 – 0.05234334224894280 i	$1.7 \cdot 10^{-7}$
8192	0.02382589337085374 – 0.05232404394907701 i	$4.4 \cdot 10^{-8}$
16384	0.0238258889851146 – 0.05231436234675194 i	$1.1 \cdot 10^{-8}$
32768	0.02382588777679670 – 0.05230951343396723 i	$2.7 \cdot 10^{-9}$
fit, $q \rightarrow \infty$	0.02382588740220057 – 0.05230465911400367 i	
horn map, Thm. 4.3	0.02382588740220057 – 0.05230465911400367 i	

TABLE 2. $\kappa(1/q)$ from the normal-form recursion (first sixteen digits shown). The corrected error decays like q^{-2} . Values for $q \geq 1024$ are certified in ball arithmetic (radius $\leq 5.4 \cdot 10^{-44}$). The fit uses the thirteen values $256 \leq q \leq 32768$ with eleven terms in $1/q$ (residual $3.6 \cdot 10^{-29}$). The last row is the horn-map value of Numerical result 4.4.

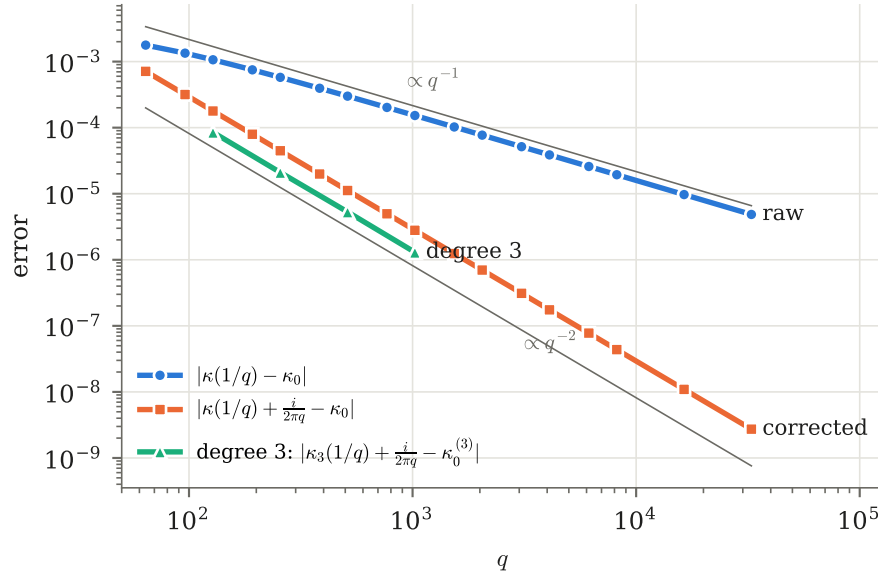


FIGURE 1. Convergence of $\kappa(1/q)$ to κ_0 in degree two and of $\kappa_3(1/q)$ to $\kappa_0^{(3)}$ in degree three (Section 8). The raw error decays like q^{-1} ; after the universal correction $-i/(2\pi q)$ it decays like q^{-2} .

Proof. With $u = 1/w$ we have $1/F = u - u^2$, $F - w - 1 = u/(1 - u)$ and $\log F - \log w = -\log(1 - u)$. Substituting gives the displayed identity. The coefficient of u^m contains c_{m-1} with the factor $-(m - 1)$ and otherwise only c_k with $k < m - 1$. The coefficient of $\log w$ is fixed by the term of order u ; it equals the résidu itératif 1 of f_0 . $\square$

Let $\Phi_{\text{att}}(w) = \lim_n (\Phi(F^n(w)) - n)$ on the attracting petal, with the principal logarithm, and let $\Psi_{\text{rep}}(Z) = \lim_n F^n(\Phi^{-1}(Z - n))$ be the repelling Fatou parametrization, with the branch $\arg w \in (0, 2\pi)$. The upper horn map $E = \Phi_{\text{att}} \circ \Psi_{\text{rep}}$ commutes with $Z \mapsto Z + 1$ and, for $\text{Im } Z$

large, has an expansion

$$E(Z) = Z + a_0 + \sum_{n \geq 1} a_n e^{2\pi i n Z}.$$

Replacing Φ_{att} or Ψ_{rep} by a translate changes a_0 , or conjugates E by $Z \mapsto Z + c$, which sends $a_n \mapsto a_n e^{2\pi i n c}$. In $W = e^{2\pi i Z}$ the horn map is the germ $W \mapsto e^{2\pi i a_0} \mathcal{P}_0(W)$, where

$$\mathcal{P}_0(W) := W \exp\left(2\pi i \sum_{n \geq 1} a_n W^n\right)$$

is the parabolic renormalization of f_0 , normalized to multiplier one [8, 10]. More generally we write $\mathcal{P}(g)$ for the germ obtained in the same way from a parabolic germ g .

Theorem 4.3. $\iota(\mathcal{P}_0) = \frac{1}{2} + \frac{a_2}{2\pi i a_1^2}$, and this quantity does not depend on the normalizations of the Fatou coordinates. The same formula holds for $\mathcal{P}(g)$.

Proof. $\mathcal{P}_0(W) = W + AW^2 + BW^3 + \dots$ with $A = 2\pi i a_1$ and $B = 2\pi i a_2 + \frac{1}{2}(2\pi i a_1)^2$, and $\iota = B/A^2$. Under $a_n \mapsto a_n e^{2\pi i n c}$ the ratio a_2/a_1^2 is unchanged, and a_0 does not enter. $\square$

Numerical result 4.4. Computing E on the line $\text{Im } Z = \frac{1}{4}$ at 60 and at 80 digits (Appendix A) gives $a_0 = 0$ to working precision and

$$\begin{aligned} \text{Re } a_1 &= -0.05895043056440687000807026251551, \\ \text{Im } a_1 &= 0.010009865898273724511535250164253, \\ \text{Re } a_2 &= 0.0012858196663089284889517464706577, \\ \text{Im } a_2 &= 0.00011738770718282661923962731179377, \end{aligned}$$

with $|a_n|$ decreasing by a factor of about 55 per step. Hence

$$\begin{aligned} \text{Re}(\iota(\mathcal{P}_0) - \tfrac{1}{2}) &= 0.023825887402200569000729189645402, \\ \text{Im}(\iota(\mathcal{P}_0) - \tfrac{1}{2}) &= -0.052304659114003666316522486207571. \end{aligned}$$

The two precision settings agree in all digits shown. This value coincides with c_0 of Numerical result 4.1 to $3.1 \cdot 10^{-28}$; other stable fits differ from it by at most $5 \cdot 10^{-25}$. We therefore claim agreement to at least 24 digits, and to 27 on the best fit.

Conjecture 4.5. The limit $\kappa_0 = \lim_{q \rightarrow \infty} \kappa(1/q)$ exists and equals $\iota(\mathcal{P}_0) - \frac{1}{2} = \frac{a_2}{2\pi i a_1^2}$. Moreover $\kappa(1/q) = \kappa_0 - \frac{i}{2\pi q} + O(q^{-2})$.

5. THE RENORMALIZATION PRINCIPLE

5.1. Heuristic. For α near 0, the near-parabolic renormalization $\mathcal{R}$ of Inou and Shishikura [8] assigns to $f_\alpha(z) = e^{2\pi i \alpha} z + z^2$ a germ $\mathcal{R}f_\alpha$ at 0 with multiplier $e^{-2\pi i/\alpha}$. By parabolic implosion [11, 16], as $\alpha \rightarrow 0$ with $-1/\alpha \rightarrow \sigma \pmod{\mathbb{Z}}$, the germ $\mathcal{R}f_\alpha$ tends to $e^{2\pi i \sigma}$ times the horn germ. For $\alpha = 1/q$ the phase is 0, and $\mathcal{R}f_{1/q}$ is a parabolic germ with multiplier one. The rotation quotient $\Phi_{1/q}$ of Corollary 3.2 is also a parabolic germ with multiplier one, obtained from f^q by passing to the quotient by the q -fold rotation. If $\Phi_{1/q}$ and $\mathcal{R}f_{1/q}$ are conformally conjugate, Corollary 3.2 and continuity of $\mathcal{R}$ give Conjecture 4.5. The same reasoning applies to the whole unfolding, and the phase can be computed exactly.

Proposition 5.1 (Kinematics). *Let $\lambda = e^{2\pi i \alpha} = \lambda_0 e^{u/q^2}$ with $\lambda_0 = e^{2\pi i p/q}$. Then $e^{-2\pi i/\alpha} = \mu e^{\tilde{u}/p^2}$, where $\mu = e^{-2\pi i q/p}$ and*

$$\tilde{u} = \frac{u}{1 + u/(2\pi i p q)} = u + \frac{i u^2}{2\pi p q} + O\left(\frac{u^3}{q^2}\right).$$

Proof. $\alpha = (2\pi ipq + u)/(2\pi iq^2)$, so $-1/\alpha + q/p = qu/(p(2\pi ipq + u))$, and $2\pi ip^2$ times this is $\tilde{u}$. $\square$

Fix $p \geq 1$ and r with $\gcd(p, r) = 1$, and let $q = pN + r$. Then $\mu = e^{-2\pi ir/p}$ does not depend on N . Let $\rho_{p,r}(u)$ be the multiplier of the p -cycle of

$$M_u(W) = \mu e^{u/p^2} \mathcal{P}_0(W)$$

that bifurcates from $W = 0$, and write $\rho_{p,r}(u) = 1 - u + \kappa_{p,r}u^2 + \rho_3u^3 + \dots$

Conjecture 5.2 (Renormalization principle). For fixed (p, r) and $q = pN + r \rightarrow \infty$,

$$R_{p/q}(u) = \rho_{p,r}(\tilde{u}(u)) + O(q^{-2})$$

coefficientwise, and locally uniformly in u on the domain of $\rho_{p,r}$.

Comparing coefficients gives the following consequences.

Corollary 5.3. *Under Conjecture 5.2,*

$$(5) \quad \kappa(p/q) = \kappa_{p,r} - \frac{i}{2\pi pq} + O(q^{-2}), \quad r_3(p/q) = \rho_3 + \frac{i \kappa_{p,r}}{\pi pq} + O(q^{-2}),$$

where, by Corollary 3.2 applied to $\mu\mathcal{P}_0$,

$$(6) \quad \kappa_{p,r} = \frac{\iota((\mu\mathcal{P}_0)^p) - \frac{1}{2}}{p}.$$

Moreover $G(p/q) \rightarrow \frac{1}{2}|u_a|$, where $\rho_{p,r}(u_a) = -1$.

In particular the first-order term $-i/(2\pi pq)$ involves no property of the germ, which explains why it is the same in degree three (Section 8). We now test the three parts of Conjecture 5.2: the limits of κ , the whole unfolding, and the first-order correction.

5.2. Bounded p . Table 3 compares (6) with bulb data for all nine classes with $2 \leq p \leq 5$. The bulb values are normal-form values of $\kappa(p/q)$ for $64 \leq N \leq 1024$, extrapolated in $1/q$. All nine agree to between $6 \cdot 10^{-9}$ and $2.6 \cdot 10^{-8}$, within the extrapolation uncertainty of about 10^{-7} . The opposite phase convention, $\mu = e^{+2\pi ir/p}$, exchanges the classes r and $p - r$ and fails. Figure 2 places these limits within the set of all $\kappa(p/q)$ at a fixed q .

5.3. The unfolding and the bulb size. For $p = 1$ the Taylor coefficients of $\rho_{1,1}$ agree with those of $R_{1/1025}$ through order eight up to the finite- q shift; for instance $r_3 = -3.78 \cdot 10^{-4} - 9.80 \cdot 10^{-4}i$ against $-3.62 \cdot 10^{-4} - 9.72 \cdot 10^{-4}i$, and $r_8 = 3.14 \cdot 10^{-9} - 4.56 \cdot 10^{-9}i$ against $3.13 \cdot 10^{-9} - 4.50 \cdot 10^{-9}i$. Since r_3 depends on the unfolding and not only on the germ, this checks that u enters as the Lavaurs phase. The shift $r_3 - \rho_3 = 1.6 \cdot 10^{-5} + 0.8 \cdot 10^{-5}i$ is the one predicted by (5), namely $1.62 \cdot 10^{-5} + 0.74 \cdot 10^{-5}i$. Table 4 compares the predicted bulb sizes with the bulbs for all ten classes with $p \leq 5$. The largest discrepancy is $9 \cdot 10^{-7}$. For $p = 1$ this gives

$$\lim_{q \rightarrow \infty} q^3 d(1/q) = 2\pi G_0, \quad G_0 = 1.02641539\dots,$$

so the $1/q$ -bulbs, which accumulate at the cusp $c = \frac{1}{4}$, have diameters $2\pi G_0 q^{-3}(1 + O(q^{-1}))$.

p	r	horn-map prediction $(\iota((\mu\mathcal{P}_0)^p) - \frac{1}{2})/p$	bulbs, extrapolated	$ \Delta $
2	1	$0.049693182903 - 0.023894096844i$	$0.049693177608 - 0.023894071809i$	$2.6 \cdot 10^{-8}$
3	1	$0.054529447219 - 0.0050171894089i$	$0.054529443083 - 0.0050171743067i$	$1.6 \cdot 10^{-8}$
3	2	$0.055036036944 - 0.023467983453i$	$0.055036033035 - 0.023467969651i$	$1.4 \cdot 10^{-8}$
4	1	$0.053737282786 + 0.0074895906470i$	$0.053737279652 + 0.0074896009583i$	$1.1 \cdot 10^{-8}$
4	3	$0.054739672309 - 0.026348641547i$	$0.054739669177 - 0.026348633046i$	$9.1 \cdot 10^{-9}$
5	1	$0.051130771685 + 0.015984969447i$	$0.051130769387 + 0.015984976959i$	$7.9 \cdot 10^{-9}$
5	2	$0.059925460063 - 0.010450443803i$	$0.059925457760 - 0.010450436333i$	$7.8 \cdot 10^{-9}$
5	3	$0.060335706660 - 0.0045365352268i$	$0.060335704038 - 0.0045365278901i$	$7.8 \cdot 10^{-9}$
5	4	$0.052624850350 - 0.029174853238i$	$0.052624847638 - 0.029174847754i$	$6.1 \cdot 10^{-9}$

TABLE 3. Bounded- p limits of κ (degree two): the prediction (6) against the bulb values extrapolated from $q = pN + r$, $64 \leq N \leq 1024$.

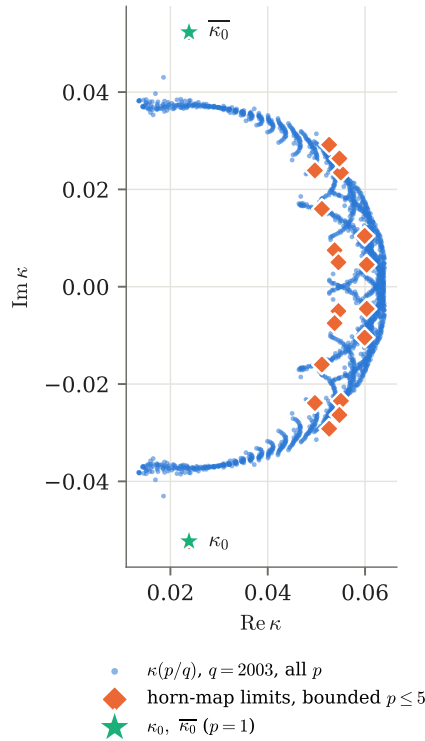


FIGURE 2. All $\kappa(p/q)$ for $q = 2003$ (values for $p > q/2$ from the symmetry $\kappa(1 - p/q) = \overline{\kappa(p/q)}$), the limits (6) for $p \leq 5$, and $\kappa_0, \overline{\kappa_0}$. The $p = 1$ bulbs are the extreme points of the arithmetic cloud.

5.4. The first-order correction. The first identity in (5) is the coefficient $c_1 = -i/(2\pi)$ of Numerical result 4.1. For $2 \leq p \leq 5$ the fitted coefficients satisfy $c_1(p) \cdot 2\pi p/(-i) = 1.0008, 1.0011, 1.0014, 1.0016$, which is within the uncertainty of the fits. A sharper test uses all Taylor coefficients at once.

Numerical result 5.4. Let $r_k(p/q)$ be the Taylor coefficients of the bulb and ρ_k those of M_u , both computed by Cauchy integrals on $|u| = 1.5$. For $2 \leq k \leq 8$, for $(p, r) = (1, 1), (2, 1), (3, 1)$ and $(3, 2)$, and for $N = 64, 128, \dots, 1024$, the residuals $|r_k(p/q) - [u^k]\rho_{p,r}(\tilde{u})|$ decrease by a

p	r	G at $q = 512p + r$	G at $q = 2048p + r$	extrapolated	horn map
1	1	1.02634395	1.02639633	1.02641535	1.02641539
2	1	1.11142282	1.11143038	1.11143322	1.11143318
3	1	1.12836657	1.12836757	1.12836781	1.12836776
3	2	1.12537076	1.12537450	1.12537590	1.12537582
4	1	1.12357397	1.12357381	1.12357368	1.12357348
4	3	1.12156857	1.12157111	1.12157204	1.12157195
5	1	1.11300416	1.11300362	1.11300313	1.11300337
5	2	1.14414695	1.14414738	1.14414703	1.14414795
5	3	1.14644523	1.14644563	1.14644539	1.14644595
5	4	1.11312789	1.11312962	1.11312984	1.11313044

TABLE 4. Bulb size G (Definition 2.4) along $q = pN + r$, extrapolated from $N = 512, 1024, 2048$, against the prediction of Corollary 5.3. The first row gives $G_0 = \lim G(1/q)$.

factor between 3.77 and 4.28 per doubling of N . This holds in all 41 cases in which the residual exceeds 10^{-8} ; below that level the round-off of the Cauchy integrals takes over. Without the substitution $u \mapsto \tilde{u}$ the factor lies between 1.1 and 2.0. In degree three, with the cubic horn germ, Cauchy integrals on $|u| = 1$ and $p = 1$, the factor lies between 3.75 and 4.06 (26 cases). Figure 3 shows the case $p = 1$.

Thus, through first order in $1/q$, the whole function $R_{p/q}$, and not only κ , is the renormalized family composed with the exact Inou–Shishikura phase. The renormalized germ itself has no $O(1/q)$ correction.

6. THE HIERARCHY AND THE CONSTANT κ_*

6.1. The iteration. For $\alpha = [0; a_1, a_2, \dots]$ the renormalization $\mathcal{R}f_\alpha$ has multiplier $e^{-2\pi i/\alpha}$. Conjugating by $z \mapsto \bar{z}$ turns its rotation number into $1/\alpha \bmod 1 = [0; a_2, a_3, \dots]$, so that $\mathcal{R}$ acts on continued fractions as the Gauss map [8]. When the first k partial quotients tend to infinity, each renormalization should therefore be close to $\mathcal{P}$ applied to the complex conjugate of the previous one. Write $\bar{G}(W) = G(\bar{W})$, normalize germs to the form $v + v^2 + O(v^3)$, and set

$$G_1 = \mathcal{P}_0, \quad G_{k+1} = \mathcal{P}(\bar{G}_k), \quad \kappa_k = \iota(G_k) - \frac{1}{2}.$$

By Theorem 4.3, $\kappa_k = a_2/(2\pi i a_1^2)$ for the horn map of $\bar{G}_{k-1}$. For $k \geq 2$ the germs G_k are given by convergent series of small radius (about 2.6 in the normalized coordinate v at $k = 2$), and the horn maps must be sampled at heights where the orbits stay well inside the disc of convergence. Table 5 lists $\kappa_1, \dots, \kappa_7$ for two independent settings of height, Fourier truncation and precision. They agree to $2 \cdot 10^{-14}$ or better at every level.

Numerical result 6.1. The differences $\Delta_k = \kappa_k - \kappa_{k-1}$ decrease geometrically: $|\Delta_{k+1}|/|\Delta_k|$ tends to 0.0419, and the ratios are $\Delta_{k+1}/\Delta_k \approx -0.01546 \mp 0.03894i$, with imaginary parts of alternating sign (Figure 4). This is the signature of an antilinear contraction $\Delta \mapsto \nu\bar{\Delta}$, as expected since the linearization of $G \mapsto \mathcal{P}(\bar{G})$ is antilinear. Fitting $\Delta_{k+1} = M\Delta_k$ with M real-linear on $\mathbb{C} \cong \mathbb{R}^2$ gives eigenvalues ± 0.041899 and the limit

$$\kappa_* := \lim_{k \rightarrow \infty} \kappa_k = 0.011929183413 - 0.037455493752i,$$

with the fits through levels 6 and 7 differing by less than $2 \cdot 10^{-12}$. Thus $\kappa_* + \frac{1}{2} = 0.5119292 - 0.0374555i$ is the cubic coefficient of the normalized fixed point $G_* = \mathcal{P}(\bar{G}_*)$.

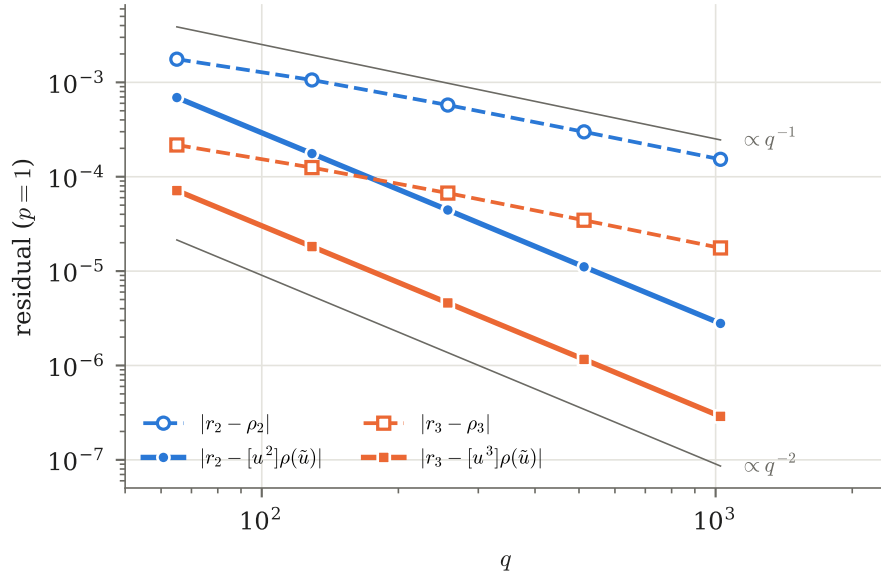


FIGURE 3. Residuals of the Taylor coefficients r_2, r_3 of $R_{1/q}$ against the renormalized family, with (solid) and without (dashed) the Möbius substitution of Proposition 5.1. The substitution removes the entire $O(1/q)$ part.

k	κ_k ($h = 1$)	$ \kappa_k - \kappa_{k-1} $	$ \kappa_k^{(h=1)} - \kappa_k^{(h=5/4)} $
1	$0.0238258874022 - 0.0523046591140 i$		—
2	$0.0112359215880 - 0.0377814219023 i$	$1.9 \cdot 10^{-2}$	$6.1 \cdot 10^{-16}$
3	$0.0119510972606 - 0.0374792854625 i$	$7.8 \cdot 10^{-4}$	$1.7 \cdot 10^{-14}$
4	$0.0119279615825 - 0.0374560782080 i$	$3.3 \cdot 10^{-5}$	$5.8 \cdot 10^{-16}$
5	$0.0119292218902 - 0.0374555354669 i$	$1.4 \cdot 10^{-6}$	$2.5 \cdot 10^{-17}$
6	$0.0119291812682 - 0.0374554947782 i$	$5.7 \cdot 10^{-8}$	$9.9 \cdot 10^{-19}$
7	$0.0119291834806 - 0.0374554938253 i$	$2.4 \cdot 10^{-9}$	$1.2 \cdot 10^{-19}$
∞	$0.011929183413 - 0.037455493752 i$		

TABLE 5. The hierarchy $\kappa_k = \iota(G_k) - \frac{1}{2}$, $G_{k+1} = \mathcal{P}(\bar{G}_k)$, computed at heights $h = 1$ (16 Fourier modes, 90 digits) and $h = \frac{5}{4}$ (14 modes, 100 digits). Level 1 is κ_0 and is common to both. The last row is the extrapolation of Numerical result 6.1.

Remark 6.2 (The holomorphic operator). Without the conjugation, $G \mapsto \mathcal{P}(G)$ is the parabolic renormalization operator whose fixed point f_* was shown to exist by Inou and Shishikura and computed by Lanford and Yampolsky [10]. They report $f_*(z) \approx z + z^2 + (0.514 - 0.0346i)z^3$ and a leading eigenvalue $\approx -0.017 + 0.040i$ of $D\mathcal{P}$ at f_* . Our code without conjugation reproduces both: the cubic coefficients converge to $0.5142768 - 0.0346276i$ (extrapolated, $\pm 10^{-7}$), and the ratio of successive differences tends to $-0.01764 + 0.04038i$. The bulbs, however, select the fixed point $G_* \neq f_*$ of the conjugated operator.

6.2. Comparison with bulbs.

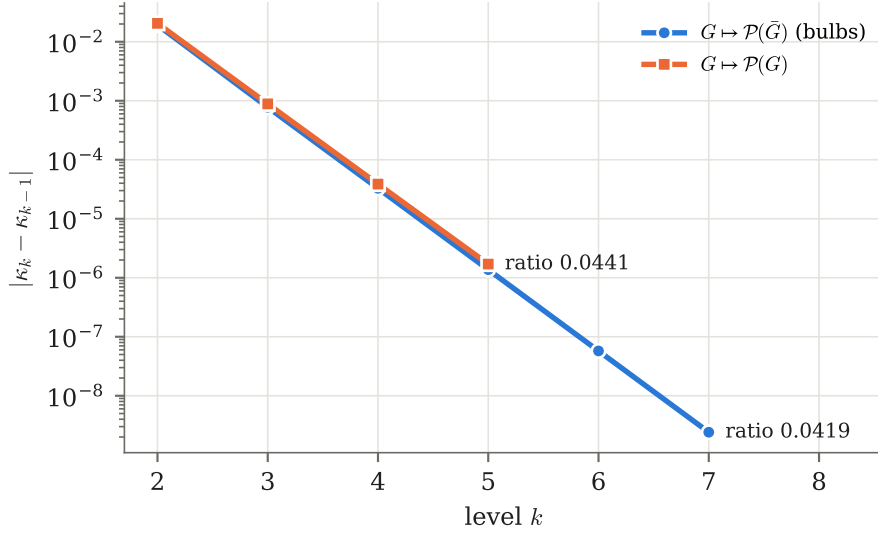


FIGURE 4. Successive differences of the two hierarchies: $G \mapsto \mathcal{P}(\bar{G})$, which governs the bulbs, and the holomorphic parabolic renormalization $G \mapsto \mathcal{P}(G)$ of Remark 6.2. The labels give the last ratio of successive differences.

Conjecture 6.3. For every $k \geq 1$,

$$\lim_{N_k \rightarrow \infty} \cdots \lim_{N_1 \rightarrow \infty} \kappa([0; N_1, \dots, N_k]) = \begin{cases} \kappa_k, & k \text{ odd,} \\ \overline{\kappa_k}, & k \text{ even.} \end{cases}$$

In particular κ_* and $\overline{\kappa}_*$ are the limits of the first correction as the number of large partial quotients of p/q grows, along odd and even k respectively.

The case $k = 1$ is Numerical result 4.4. For $k = 2$, formula (6) with $r = 1$ gives the limit of $\kappa([0; N, p])$ as $N \rightarrow \infty$ as an index of $(e^{-2\pi i/p} \mathcal{P}_0)^p$. We computed this index for $p \leq 256$ with the normal form of Theorem 3.1 generalized to arbitrary germs, an $O(p^3)$ recursion, and extrapolated in $1/p$. The result, $0.0112359330 + 0.0377814233i$, equals $\overline{\kappa}_2$ to $1.2 \cdot 10^{-8}$. The conjugate appears because $p^{-1} \bmod q \rightarrow 0^-$ along this sequence, against 0^+ for $p = 1$. Without the conjugation in the definition of G_2 one obtains $0.014770 - 0.033911i$, which the data exclude.

We also tested the diagonal sequences $[0; N, \dots, N]$, computing κ with the normal form at 128 and 192 bits (the two agree to $4 \cdot 10^{-35}$) and fitting polynomials in $1/N$. For $k = 2$ and $16 \leq N \leq 160$ ($q \leq 25601$) the limit agrees with $\overline{\kappa}_2$ to $3 \cdot 10^{-10}$, so the diagonal and the iterated limits coincide. For $k = 3$ and $6 \leq N \leq 40$ ($q \leq 64080$) the limit is $0.0119509937 - 0.0374804294i$. It agrees with κ_3 to $1.2 \cdot 10^{-6}$, and every fit with $N \geq 10$ and five or six terms lies within $1.4 \cdot 10^{-5}$ of κ_3 . For comparison, $|\kappa_3 - \kappa_2| = 7.8 \cdot 10^{-4}$ and $|\kappa_3 - \overline{\kappa}_3| = 7.5 \cdot 10^{-2}$. Thus Conjecture 6.3, including its parity rule, holds numerically for $k \leq 3$. Level 4, where $|\kappa_4 - \kappa_3| = 3.3 \cdot 10^{-5}$, would require periods $q \approx N^4$ with $N \gtrsim 30$.

7. COMPONENT DEPENDENCE AND THE $\frac{1}{2}$ -ROOT

7.1. The period-two disc. Remark 2.2 applies to the satellites of every hyperbolic component. For the period-two disc $W_{1/2}$, $c = -1 + \lambda/4$, with λ the multiplier of the attracting

2-cycle, the values of κ at generic p/q are close to those of the cardioid at the same p/q [15], but the bounded- p limits differ. The renormalization picture attributes these limits to the parabolic germ at the root of the component, here $c = -\frac{3}{4}$. This point is also the root of the cardioid's own $\frac{1}{2}$ -satellite, which the cardioid approaches along $[0; 1, 1, m, \dots]$ with $m \rightarrow \infty$. This suggests the following.

Conjecture 7.1 (Root principle). The limits of κ and of G along satellites $[0; N, t]$ of a hyperbolic component, as $N \rightarrow \infty$ with the tail t fixed, depend only on the parabolic germ at the root of the component. In particular the satellite $[0; N, t]$ of $W_{1/2}$ and the satellite $[0; 1, 1, N - 1, t]$ of the cardioid have the same limits.

Table 6 confirms the second statement for three tails, and for the bulb size, to about 10^{-9} . The disc values come from Cauchy integrals in the multiplier of the 2-cycle, and the cardioid values from the normal form. When the first partial quotient stays bounded the correspondence fails: at $N = 256$ the two sides differ by between $3 \cdot 10^{-5}$ and $1.5 \cdot 10^{-3}$ for the prefixes $(2, 9, 4)$, $(3, 5)$, $(1, 4)$, $(5, 1, 2)$ and (7) , and the disc is then closer to the cardioid at the same p/q . So only the first renormalization, and hence the germ at the root, distinguishes components.

7.2. The horn map of $-z + z^2$. At $c = -\frac{3}{4}$ the map is conjugate to $f(z) = -z + z^2$, and $g = f \circ f = z - 2z^3 + z^4$ has a triple fixed point at 0. Its attracting petals lie along $\pm\mathbb{R}$ and its repelling petals along $\pm i\mathbb{R}$, and f exchanges the members of each pair.

Lemma 7.2 (Two-petal Fatou coordinate). *There is a unique formal series*

$$\Phi(z) = \frac{1}{4}z^{-2} + d_{-1}z^{-1} + \beta \log z + \sum_{k \geq 1} d_k z^k$$

with $\Phi \circ g = \Phi + 1$. Its coefficients are rational: $d_{-1} = \frac{1}{4}$, $\beta = \frac{11}{8}$, $d_1 = -\frac{5}{16}$, $d_2 = \frac{75}{64}$, ... The coefficient β is the résidu itératif $\frac{3}{2} - \iota(g)$ of g , and $\iota(g) = \iota_{1/2} = \frac{1}{8}$.

Proof. Write $g(z) = z(1 + h)$ with $h = -2z^2 + z^3$. Then $\Phi \circ g - \Phi = \sum_k d_k z^k ((1 + h)^k - 1) + \beta \log(1 + h)$, and $z^k((1 + h)^k - 1) = -2kz^{k+2} + O(z^{k+3})$. The coefficient of z^m therefore contains d_{m-2} with the factor $-2(m-2)$ for $m \neq 2$, and β with the factor -2 for $m = 2$, together with unknowns of lower index. Solving order by order gives uniqueness, with the constant term d_0 free, and the stated values. The index is $\text{Res } dz/(2z^3 - z^4) = \frac{1}{8}$, which agrees with $\iota_{1/2}$ in Remark 2.3. $\square$

Consider orbits leaving the repelling petal $+i\mathbb{R}$ above the height $\text{Im } Z = \beta\pi/2 \approx 2.16$ in the Fatou coordinate. They reach the attracting petal $-\mathbb{R}$, and f , a half step of g , carries them to $+\mathbb{R}$. This defines an upper horn map $E_{1/2}(Z) = Z + a_0 + \sum a_n e^{2\pi i n Z}$ on the cylinder of f -orbits, and a parabolic germ $\mathcal{P}_{1/2}$ as in Section 4.

Numerical result 7.3. Computed at the heights 4.5 and 5.5 with 100 digits, which agree in all digits shown,

$$\begin{aligned} \text{Re } \kappa_{1/2} &= 0.018405261617964781008961965439681, \\ \text{Im } \kappa_{1/2} &= -0.042903723880230475207353337798906, \end{aligned}$$

where $\kappa_{1/2} := \iota(\mathcal{P}_{1/2}) - \frac{1}{2}$. This agrees with the bulb limits of Table 6 to $3 \cdot 10^{-10}$ for the disc and $2 \cdot 10^{-9}$ for the cardioid. The germ $\mathcal{P}_{1/2}$ also carries the rest of the theory. With the phase $e^{-2\pi i r/p}$ and $r = 1$ (since $[0; N, t] = t/(tN + 1)$), formula (6) gives $0.04755255819 - 0.01974956104i$ for $t = 2$ and $0.05330701191 - 0.00260595580i$ for $t = 3$. The unfolding $e^u \mathcal{P}_{1/2}$ gives $G = 1.0201625875$ for $p = 1$ and $G = 1.1070111364$ for $p = 2$, against 1.107011147 for disc bulbs along $2/(2N + 1)$.

	disc $[0; N, t]$	cardioid $[0; 1, 1, N - 1, t]$	$ \Delta $
$\kappa, t = \emptyset$	$0.01840526173 - 0.04290372357i$	$0.01840526170 - 0.04290372213i$	$1.4 \cdot 10^{-9}$
$\kappa, t = 2$	$0.04755255860 - 0.01974956083i$	$0.04755255811 - 0.01974956031i$	$7.1 \cdot 10^{-10}$
$\kappa, t = 3$	$0.05330701166 - 0.002605955494i$	$0.05330701184 - 0.002605955354i$	$2.2 \cdot 10^{-10}$
$G, t = \emptyset$	1.020162587	1.020162587	$5.3 \cdot 10^{-10}$

TABLE 6. The period-two disc near its root against the cardioid near its $\frac{1}{2}$ -root. The disc column uses $[0; N, t]$ in the multiplier of the 2-cycle, the cardioid column $[0; 1, 1, N - 1, t]$. Both are extrapolated from $N = 128, \dots, 2048$ with three terms in $1/q$.

The first-order coefficient of the disc, $\kappa_{W_{1/2}}(1/q) = \kappa_{1/2} + (-0.17743 - 0.19381i)/q + O(q^{-2})$, is not the kinematic value $-i/(2\pi q)$ of Corollary 5.3. This is not a contradiction: at the degenerate parabolic point $c = -\frac{3}{4}$ the multiplier of the 2-cycle is not the Inou–Shishikura rotation, so Proposition 5.1 does not apply to it.

8. DEGREE THREE: A SECOND UNIVERSALITY CLASS

For $z^3 + c$ the main component is parametrized by the multiplier λ of the fixed point $z = (\lambda/3)^{1/2}$, with $c = z - z^3$. Its cusp at $\lambda = 1$, where $z_* = 1/\sqrt{3}$ and $c_* = 2/(3\sqrt{3})$, has the parabolic germ $w \mapsto w + \sqrt{3}w^2 + w^3$. In $v = \sqrt{3}w$ this is

$$g(v) = v + v^2 + \frac{1}{3}v^3,$$

whose résidu itératif is $1 - \frac{1}{3} = \frac{2}{3}$. Lemma 4.2 generalizes with $F(w) = -1/g(-1/w)$ and $\Phi(w) = w - \frac{2}{3}\log w + \sum c_k w^{-k}$, where $c_1, c_2, \dots = \frac{2}{9}, \frac{5}{54}, \frac{14}{243}, \frac{89}{2430}, \dots$. The horn map gives, at 50 and at 70 digits,

$$\kappa_0^{(3)} := \frac{a_2}{2\pi i a_1^2} = 0.143599278095291569488064104088 - 0.030311647898502537611943110858i.$$

Table 7 shows that $\kappa_3(1/q) + \frac{i}{2\pi q}$ converges to this value like q^{-2} . The correction $-i/(2\pi q)$ is thus the same as in degree two, as Corollary 5.3 predicts, while the constant is not. The bounded- p predictions hold in degree three as well: κ agrees to within $5 \cdot 10^{-6}$ for $(p, r) = (1, 1), (2, 1), (3, 1), (3, 2)$ at $q \approx 512p$.

The bulb size needs one more observation.

Proposition 8.1. *Let W be a bounded hyperbolic component of $z^d + c$. Its multiplier map $\rho : W \rightarrow \mathbb{D}$ has the form $\rho = \phi^{d-1}$ with $\phi : W \rightarrow \mathbb{D}$ a conformal isomorphism. In particular $\rho = -1$ at exactly $d - 1$ points of ∂W .*

Proof. The map ρ is proper, since the cycle becomes indifferent at ∂W . Its zeros are the parameters at which the critical point lies on the cycle; there is exactly one, the centre, because W is a cell with a unique centre [13]. At the centre c_0 we have $\rho = \prod_j d z_j^{d-1}$ with one factor $z_j = f_c^n(0)$ vanishing. Moreover c_0 is a simple zero of $P(c) = f_c^n(0)$: P has integer coefficients and $P' \equiv 1 \pmod{d}$ by induction, so $P'(c_0) \neq 0$ at the algebraic integer c_0 (Gleason’s argument). Hence ρ vanishes to order exactly $d - 1$ at c_0 . A proper holomorphic map from a disc to $\mathbb{D}$ with a single zero of order $d - 1$ is a Blaschke product ϕ^{d-1} with ϕ a conformal isomorphism. $\square$

In degree three, then, there are two points with multiplier -1 , and both are predicted. At $p = 3, r = 1$, the horn map gives $G = 1.196074$ and 1.227092 , against the extrapolated bulb values 1.196075 and 1.227091 .

q	$\kappa_3(1/q)$	$ \kappa_3(1/q) + \frac{i}{2\pi q} - \kappa_0^{(3)} $
128	$0.1436180525 - 0.03147252376 i$	$8.5 \cdot 10^{-5}$
256	$0.1436043106 - 0.03091278633 i$	$2.1 \cdot 10^{-5}$
512	$0.1436005784 - 0.03061736685 i$	$5.3 \cdot 10^{-6}$
1024	$0.1435996089 - 0.03046578967 i$	$1.3 \cdot 10^{-6}$
horn map of $v + v^2 + \frac{1}{3}v^3$		$0.1435992781 - 0.03031164790 i$

TABLE 7. Degree three: $\kappa_3(1/q)$ from Cauchy integrals on $|u| = 1$ (Appendix A) against the horn map of the cubic cusp germ.

9. ARITHMETIC NATURE

For each of $\kappa_0, \kappa_0^{(3)}$ and $\kappa_{1/2}$ we searched with PSLQ [4], at 28 digits and coefficient bound 1000, for integer relations between the real part, the imaginary part, the squared modulus or the argument divided by π , and every subset of at most three of $\{1, \pi, \pi^2, 1/\pi, \log 2, \zeta(3), G, \gamma\}$, where G is Catalan's constant. For κ_0 we also tried $\text{Re}(2\pi i \kappa_0)$, $\text{Im}(2\pi i \kappa_0)$, $|a_1|$ and $\arg a_1/\pi$. No relation was found. This is consistent with the general expectation that invariants of parabolic germs, such as the Écalle–Voronin moduli [3, 17], are transcendental numbers without closed form. We regard $\kappa_0, \kappa_0^{(3)}$ and $\kappa_{1/2}$ as new constants of this kind.

10. DISCUSSION

The first-order theory of the satellite bulbs is the Ford-circle picture, with radii q^{-2} and Farey tangencies. The first correction, $\kappa(p/q)$, is the holomorphic index of a parabolic germ (Theorem 2.1), and it equals the index of the rotation quotient minus $\frac{1}{2}$ (Corollary 3.2). Numerically it is controlled by near-parabolic renormalization. The constant κ_0 comes from the simplest instance, $\alpha = 1/q \rightarrow 0$. Bounded p corresponds to one renormalization step with phase $e^{-2\pi i r/p}$, and the first order in $1/q$ comes from the exact Möbius phase. Large partial quotients correspond to iterated renormalization, which contracts by a factor 0.042 per step towards a fixed point that differs from the classical one. Hyperbolicity of $\mathcal{R}$ would explain why $\kappa(p/q)$ depends mainly on the last partial quotients, that is, on $p^{-1} \bmod q$ [15]. Finally, the component enters only through the germ at its root. Farey-type combinatorics survives at every order, but the metric data at a rational are those of a parabolic germ, and they are transcendental.

Open problems.

- (1) Prove Conjecture 4.5, for instance by establishing a conformal conjugacy between $\Phi_{1/q}$ and $\mathcal{R}f_{1/q}$ together with continuity of $\mathcal{R}$ at integer $1/\alpha$.
- (2) Prove Conjecture 5.2, in particular that the renormalized germ has no $O(1/q)$ correction beyond the phase of Proposition 5.1. By Corollary 5.3 this gives the universal term $-i/(2\pi pq)$. Identify the $O(q^{-2})$ term.
- (3) Prove Conjecture 6.3, and describe the fixed point G_* of $G \mapsto \mathcal{P}(\bar{G})$ and its spectrum.
- (4) Compute the generic limit function $\kappa(p/q)$, $p \rightarrow \infty$, by iterated renormalization with bounded partial quotients, including its values at quadratic irrationals as periodic points of $\mathcal{R}$.
- (5) Prove Conjecture 7.1. Derive the first-order coefficient $-0.17743 - 0.19381 i$ of the disc from the kinematics of the degenerate parabolic point $c = -\frac{3}{4}$, and treat the roots of other components in the same way.

- (6) Decide whether κ_0 can be expressed in closed form through the Écalle–Voronin invariants of $z + z^2$.

APPENDIX A. NUMERICAL METHODS AND ERROR CONTROL

A.1 Normal form. Recursion (4) was run in `mpmath` [14] at 50 working digits for $64 \leq q \leq 8192$ ($p = 1$) and $q \leq 8193$ (bounded p). It was also implemented in Arb ball arithmetic [9]. There the online convolution $(H^2)_k$ is evaluated in blocks: completed blocks are folded in by truncated series products, and the remainder is summed directly. The working precision is doubled until the certified radius of κ is below 10^{-40} . For $p = 1$ the radii are $1.6 \cdot 10^{-50}$ at $q = 1024$ and below $2 \cdot 10^{-111}$ at $q = 32768$, and the two implementations agree to the 40 digits stored. The ball version is 60–110 times faster; $q = 32768$ takes about two minutes. For generic p the coefficients H_k are very large and ball radii, which cannot see cancellation, grow until they carry no information. For those values we ran the same code in midpoint arithmetic at two precisions, 128 and 192 bits, and took their difference as the error; it never exceeded $4 \cdot 10^{-35}$. The recursion for general germs used in Section 6 was run in `mpmath` at 40–60 digits.

A.2 Extrapolation. Limits in q or N come from least-squares fits in powers of $1/q$ or $1/N$, solved by QR decomposition at 40–80 digits. Each is reported only when it is stable, to the stated precision, under changes of the fit order and of the smallest q used. The limit of the hierarchy (Numerical result 6.1) uses a real-linear model $\Delta_{k+1} = M\Delta_k$ fitted to the last three differences. This model is exact for both linear and antilinear geometric contractions, whereas componentwise Aitken extrapolation is not exact for antilinear ones.

A.3 Horn maps. For the cusp germs the Fatou series were truncated at $K = 40$ terms and evaluated at $|w| \geq 800$, where the truncation error is below 10^{-60} ; the coefficients $|c_k|$ grow roughly like $k!/(2\pi)^k$. We used $n = 1000$ – 1800 repelling iterations and computed Fourier coefficients from 64–80 samples on $\text{Im } Z = \frac{1}{4}$, with aliasing below 10^{-58} . For the germs G_k of the hierarchy, which are known only as convergent series, the horn maps were sampled at $\text{Im } Z = 1$ and $\frac{5}{4}$ so that all orbits stay well inside the disc of convergence. A first run at $\text{Im } Z = \frac{1}{2}$ was inaccurate at the 10^{-3} level and was discarded. For the two-petal germ of Section 7 the exact series of Lemma 7.2 was truncated at $K = 40$ – 60 , with 900–1000 iterations of g and samples at $\text{Im } Z = 4.5$ – 5.5 , well above the centre $\beta\pi/2$ of the repelling cylinder. A Fourier coefficient a_n extracted at height h carries noise amplified by $e^{2\pi nh}$, and the working precision (60–160 digits) was chosen accordingly. Independent heights agree to all digits reported.

A.4 Multiplier data. The Taylor coefficients r_k were computed by Cauchy integrals of $R_{p/q}$ over a circle, continuing the q -cycle numerically, and those of the renormalized families in the same way. For $z^2 + c$ radii up to $|u| = 2.5$ are admissible. For $z^3 + c$ the function $R_{p/q}$ has a singularity inside $|u| < 2.5$, and the radius must be at most 1.5. The condition $r_1 = -1$ of Theorem 2.1 is a reliable diagnostic: on the larger circle it fails, with $r_1 = -1.032$. In degree three each satellite has two antipodes (Proposition 8.1), and both were continued in the target multiplier from 0 to -1 .

A.5 Reproducibility. All code, data and tests are in the accompanying repository [15]. The package `bulbford` contains the normal forms (`normal_form`, `normal_form_ball`), the horn maps (`horn`, `horn_half`), the germ and renormalization tools (`germ`, `renorm`), the extrapolation (`extrapolate`) and the dynamics (`dynamics`, whose cycle iteration is compiled with Numba). Every table and figure is generated from the data files by `paper/tables.py` and `paper/figures.py`. The test suite checks Theorems 2.1 and 3.1, the value of κ_0 to 25 digits, the bounded- p , kinematic, hierarchy and degree-three predictions, and the constant $\kappa_{1/2}$.

Data availability. All data supporting the results are contained in the repository [15].

Acknowledgments. The computations and the drafting of this paper were carried out with the assistance of an AI system (Claude, Anthropic).

REFERENCES

- [1] X. Buff and A. Epstein, *A parabolic Pommerenke–Levin–Yoccoz inequality*, Fund. Math. **172** (2002), 249–289.
- [2] A. Douady and J. H. Hubbard, *Étude dynamique des polynômes complexes*, Parties I et II, Publications Mathématiques d’Orsay 84-02 and 85-04, 1984–1985.
- [3] J. Écalle, *Les fonctions résurgentes*, Tomes I–III, Publications Mathématiques d’Orsay, 1981–1985.
- [4] H. R. P. Ferguson, D. H. Bailey and S. Arno, *Analysis of PSLQ, an integer relation finding algorithm*, Math. Comp. **68** (1999), 351–369.
- [5] A. C. Fowler and M. J. McGuinness, *The size of Mandelbrot bulbs*, Chaos Solitons Fractals X **3** (2019), 100019.
- [6] L. R. Ford, *Fractions*, Amer. Math. Monthly **45** (1938), 586–601.
- [7] J. Guckenheimer and R. McGehee, *A proof of the Mandelbrot N^2 conjecture*, Report No. 15, Institut Mittag-Leffler, 1984.
- [8] H. Inou and M. Shishikura, *The renormalization for parabolic fixed points and their perturbation*, preprint; see <https://www.math.kyoto-u.ac.jp/~mitsu/pararenorm/>.
- [9] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), 1281–1292.
- [10] O. E. Lanford III and M. Yampolsky, *Fixed Point of the Parabolic Renormalization Operator*, SpringerBriefs in Mathematics, Springer, 2014; arXiv:1108.2801.
- [11] P. Lavaurs, *Systèmes dynamiques holomorphes: explosion de points périodiques paraboliques*, Thèse, Université Paris-Sud, Orsay, 1989.
- [12] J. Milnor, *Dynamics in One Complex Variable*, 3rd ed., Annals of Mathematics Studies 160, Princeton University Press, 2006.
- [13] J. Milnor, *Hyperbolic components* (with an appendix by A. Poirier), in: Conformal Dynamics and Hyperbolic Geometry, Contemp. Math. **573**, Amer. Math. Soc., 2012, 183–232.
- [14] The mpmath development team, *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, <https://mpmath.org>.
- [15] *mandelbrot-bulbs-and-ford-circles-research*, code and data repository accompanying this paper, including the research register RESEARCH_bulb-ford-correction.md.
- [16] M. Shishikura, *Bifurcation of parabolic fixed points*, in: The Mandelbrot Set, Theme and Variations (Tan Lei, ed.), London Math. Soc. Lecture Note Ser. 274, Cambridge Univ. Press, 2000, 325–363.
- [17] S. M. Voronin, *Analytic classification of germs of conformal mappings $(\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0)$ with identity linear part*, Funct. Anal. Appl. **15** (1981), 1–13.